School of Computing and Creative Technologies

Assessment Brief

Submission details

Module title:	Distribute & Enterprise Software Development
Module code:	UFCFTR-30-3
Module Leader:	Dr. Khoa Phung, Dilshan Jayatilake
Assessment title:	Group Project with Demonstration
Year:	2025-26
Assessment type:	Project, In-class Demonstration
Assessment weighting:	60% of total module mark
Size or length of assessment:	N/A
Module learning outcomes assessed by this task:	MO2. Evaluate the legal, ethical, social and professional implications in development life-cycle of large scale software systems. MO3. Apply a current industrial project management approach when undertaking a large-scale software development project. MO4. Identify security issues in distributed or enterprise level software systems in order to implement preventive measures.
Use of AI in assessment:	Generative AI cannot be used for authoring the source code and report(s). If this has been done, it must be declared and acknowledged.
Guidance on Referencing (inc AI):	Please note that the aim of referencing is to demonstrate you have read and understood a range of sources to evidence your key points. You need to list the references consistently and in such a way as to ensure the reader can follow up on the sources for themselves. Referencing - Study skills | UWE Bristol Using generative AI at UWE Bristol - Study skills | UWE Bristol

Submission and feedback dates

Submission deadline:	Before 14:00 on 7th May 2026 (*not eligible for 48 hours late submission window*). All times are 24-hour clock, current local time (at time of submission) in the UK.
Submission format:	Source Code (via GitHub or GitLab), Contributions Matrix, Presentations
Marks and Feedback due on:	27th May 2026
Marks and Feedback will be provided via:	Blackboard, feedback form.

Completing your assessment

SUBMISSION CHECKLIST

Throughout the Project (No Written Reports Are Required)

Weekly Standups:

- Attend weekly standup sessions (individually assessed, best 6 scores counted).
- Provide 2-5-minute progress report each time.

Sprint 0 (First 3 weeks of Term 2):

- Non-assessed sprint. No marks will be given.
- This sprint focuses on group forming and technical preparations.

Sprint 1 & 2 Reviews:

- Attend the scheduled sprint review (end of each sprint) sessions with tutor.
- Provide 10-minute live/recorded demonstration of the developed features planned for the sprint.
- Report progress and any challenges encountered.

Final Sprint 3 Review:

- Attend the scheduled presentation session.
- Demonstrate the developed system functionality.
- Show how the system passed the provided test cases.
- Demonstrate technical understandings.

Final (Required) Submissions (End of Sprint 3 – 7th May 2026):

- GitHub/GitLab repository link – *ensure repository is **publicly** accessible*.
- **Signed** contributions matrix – agreed upon and signed by all group members.
    - Note: *If the contributions matrix is not signed or submitted, we will consider that all the group members have equally contributed to the project. The contributions matrix could be challenged based on tutor's observations.*

GitHub/GitLab Repository Includes:

- Complete source code with Docker containerisation.
- README file with setup and running instructions.
- Clear commit history showing individual contributions.

Important: No other written reports, documentation (except the signed contribution matrix), or project proposals are required as separate submissions.

What am I required to do on this assessment?
The case study used in this assessment for software development is the same one as Assessment Task 1. The case study has been provided and is available on Blackboard.

This group-based assessment develops technical expertise and collaborative skills through modern project management methodology – Scrum. Each group must have ***4 to 5 members, with the maximum being 5***.

The assessment focuses on large-scale software development including component development, external services, security, containerisation, and teamwork performance.

Where should I start?
Students are expected to commence this coursework by meeting with group members for this assessment. Students should consider group formation carefully, taking into account complementary skills and commitment levels rather than simply choosing close friends. **Groups must be formed before the start of Sprint 1 (16th February 2026). Once Sprint 1 starts, groups cannot be changed.**

This approach will better prepare you for professional environments where effective collaboration depends on diverse expertise and shared dedication to project goals.

Note: *Students who are unable to find suitable group members should contact the module leaders for assistance with group allocation within Sprint 0. If students are not in groups by the end of Sprint 0, the module team will allocate these students to random groups (either new groups or existing groups with less than 5 members).*

Step-by-Step Process

1. Initial Group Meeting and Preparation

- Contact all group members within the first week and arrange initial meeting.
- Establish communication channels and regular meeting times.
- Set up shared document storage and version control access.
- *If any group member becomes unresponsive or fails to participate actively after the first two weeks, please contact the module leaders immediately to discuss potential workload/requirements adjustments.*

2. Requirements Analysis and Scope

- *All groups are expected to implement the complete case study* – not just individual Term 1 components.
- *Test cases will be provided at the start of Sprint 1 – final system is (ideally) expected to pass all the test cases.*
- Use Term 1 BPMN diagrams and business logic as foundation, but extend/refine for technical implementation.
- [Optional] Connect Term 1 work to Term 2 implementation via APIs and/or containerisation where applicable.

3. Project Management

- Break down complete case study into measurable tasks with clear acceptance criteria.
- Select one professional platform (Jira, Trello, GitHub Projects, etc.) for consistent use.
- Create timeline with dependencies and milestones for all requirements.

4. Development Phase

Build complete system using Docker and Django/Django Rest Framework across **three 3-week sprints**:

- **Sprint 1 (15% of the Group Mark):** Core architecture and basic functionality.
- **Sprint 2 (15% of the Group Mark):** Expanded features and integration.
- **Sprint 3 (50% of the Group Mark):** Complete implementation and refinement. Demonstrate how the system passes the provided test cases and technical understandings about the developed system.

5. Contributions Matrix and Group Issues

- Document individual contributions from Sprint 1 onwards.
- Track contributions via version control and project management tools.
- **For non-contributing members:**
    - Document all engagement attempts.
    - Contact module leaders as soon as possible.
    - Complete signed contributions matrix at the end of Sprint 3.

6. Weekly Standups

- Attend weekly standups throughout all sprints for progress monitoring and team coordination (best 6 scores counted).
- Provide individual progress reports covering: work completed, planned work, and impediments (approximately two minutes per group member).
- **Individual Assessment:** Standup attendance and participation are assessed individually and directly affect personal marks regardless of group performance.

How are individual marks calculated?
Each student's final mark has two components:

1. **Group Project Work (80%)**

Individual Project Mark = Scaling Factor × Group Mark

- Where:
- **Group Mark = (15% × Sprint 1) + (15% × Sprint 2) + (50% × Sprint 3)**
- **Scaling Factor = Individual Contribution Percentage / Highest Contribution Percentage in Group**

Groups must complete the **Contributions Matrix Template.docx** documenting each member's percentage contribution across all project components.

Example:

Student	Contribution %	Calculation (Ratio)	Scaling Factor	Individual Mark
Student A	40% (Max)	40 / 40	1.0	**70** (100% of Group Mark)
Student B	30%	30 / 40	0.75	**52.5** (75% of Group Mark)

2. **Individual Standup Performance (20%)**

Individual Standup Mark = (Sum of individual's highest 6 standup scores ÷ 30) × 100

- **9 standups total** across three sprints
- **Best 6 scores counted** (lowest 3 discarded)

Standup Scoring:

- **5** - Excellent attendance, clear detailed reports, proactive
- **4** - Good attendance, adequate reports, engaged
- **3** - Satisfactory attendance, basic reports
- **2** - Poor reports, limited engagement
- **1** - Attends but minimal contribution
- **0** - Non-attendance

Final Individual Mark = (Individual Project Mark × 0.80) + (Individual Standup Mark × 0.20)

What do I need to do to pass?
Reasonable contribution to the group project that demonstrates:

- **Functional Requirements:** System implements core case study features and passes essential test cases (minimum 40% of the provided test cases).
- **Technical Standards:** Working Docker containerisation, basic Django/DRF implementation, and functional API endpoints.
- **Individual Contribution:** Documented evidence of meaningful participation through version control commits and sprint deliverables.
- **Group Collaboration:** Active participation in group meetings, sprint reviews, and completion of signed contributions matrix.
- **Weekly Standups:** Regularly attend weekly standups (best 6 standup scores are counted).

How do I achieve high marks in this assessment?
Equal contribution to the teamwork that produced a working product delivers all requirements in a very/excellent/outstanding quality as described in making scheme.

How does the learning and teaching relate to the assessment?
All lectures and practical sessions are very relevant to this assessment.

Students need to beware of that the final demonstration of the developed software will include Q/A to group members to assess the level of comprehension and understanding.

What additional resources may help me complete this assessment?
All required resources will be supplied via BB.

What do I do if I am concerned about completing this assessment?
It is recommended that you review all of the relevant materials on Blackboard. You can also speak to your module leader for advice and guidance.

UWE Bristol offer a range of Assessment Support Options that you can explore through this link, and both Student Support Advisers and Wellbeing Support are available.
For further information, please see the Student study essentials.

How do I avoid an Assessment Offence on this module?
Use the support above if you feel unable to submit your own work for this module.

This is a practice-based assessment, no AO is expected unless the same project is submitted by multiple groups.

Marks and Feedback

Your assessment will be marked according to the following marking criteria.
You can use these to evaluate your own work before you submit.

When submitting your work, you will be required to confirm that the work is your own, and text-matching software and other methods are routinely used to check submissions against other submissions to the university and internet sources. Details of what constitutes plagiarism and how to avoid it can be found on UWE's Study Skills pages about avoiding plagiarism.

Marking Criteria – Sprint 1 Review (15% of the Group Mark)

Criteria	Fail (<40)	Pass (40-49)	2nd Class (50-69)	1st Class (70+)
Project Management & Planning (50%) **Relates to MO2 and MO3**	Unsuccessful attempt or no evidence of project management planning. Obvious lack of knowledge in adapting Term 1 work to development context. Misunderstanding of sprint methodology fundamentals.	Successfully adapted Term 1 requirements into basic development tasks. Several tasks roughly identified and vaguely assigned to members. Basic Gantt chart or task tracking system in place. Product backlog roughly established from existing requirements.	Successfully detailed task breakdown from Term 1 requirements. Reasonable project management plan with clear task assignments. All team members have defined roles and responsibilities. Delivery schedule established with realistic milestones. Evidence of progress tracking and issue identification.	Optimal adaptation of Term 1 work into comprehensive development plan. Excellent project management with detailed task descriptions and optimal assignments. Clear delivery schedule with contingency planning. Evidence of proactive issue resolution and team reflection. Professional presentation demonstrating strong teamwork insights.
Technical Development & Implementation (50%) **Relates to MO3 and MO4**	No or unsuccessful implementation of core system components. Misunderstanding or unawareness of Django/Docker fundamentals. No functional features demonstrated.	Basic system architecture established. Database connection achieved with simple Model classes. Basic user authentication functionality implemented. Containerisation attempted with limited success.	Successful database integration with Django models including CRUD operations. Complete user authentication system (login/logout) functional. System successfully containerised using Docker. Clear architecture foundation for remaining development phases.	Comfortable and optimal use of database operations with well-designed model classes. Robust authentication system integrated with user interface. Optimal containerisation with proper configuration management. Additional core features implemented beyond minimum requirements.

				Clear technical understanding demonstrated with wider context awareness.

Marking Criteria – Sprint 2 Review (15% of the Group Mark)

Criteria	Fail (<40)	Pass (40-49)	2nd Class (50-69)	1st Class (70+)
Project Management & Planning (50%) **Relates to MO2 and MO3**	Unsuccessful attempt or no evidence of updated project management planning. Obvious lack of knowledge in progressing from Sprint 1. Misunderstanding of iterative development fundamentals.	Product backlog updated with Sprint 2 requirements. Several tasks identified and assigned based on team roles. Basic progress tracking from Sprint 1 deliverables. Some evidence of planning adjustments.	Detailed role identification with clear task mapping. Product backlog reasonably updated with Sprint 2 scope. All tasks rigorously identified and assigned. Realistic delivery schedule established.	Optimal role identification and task allocation. Excellent product backlog management with clear Sprint 2 updates. Comprehensive delivery schedule with contingency planning. Evidence of proactive project management.
Technical Development & Implementation (50%) **Relates to MO3 and MO4**	Unsuccessful or very limited delivery of planned Sprint 2 tasks. No evidence of meaningful progress from Sprint 1. System functionality remains basic with no advancement.	Few additional model, view, and template classes delivered. System architecture becoming clearer. Evidence of several partially completed features. Some new functionality beyond Sprint 1 baseline.	Successful delivery of majority Sprint 2 tasks including backend and frontend components. Clear containerised architecture with improved structure. Further development tasks identified with few outstanding items. Some work-in-progress tasks slightly behind schedule.	Successful delivery of majority or almost all planned Sprint 2 features. Optimal system architecture with advanced containerisation. Integration of external services and security measures planned or initiated. Few remaining tasks focused on refinement and enhancement. Clear preparation for Sprint 3 features.

Marking Criteria – Sprint 3 Review (Final Presentation) (50% of the Group Mark)

Criteria	Fail (<40)	Pass (40-49)	2nd Class (50-69)	1st Class (70+)
Test Case Implementation and Technical Understanding (50%) **Relates to MO3 and MO4**	Unsuccessful attempt or no implementation of provided test cases. System fails majority of essential test cases with no clear understanding of failures. Obvious lack of technical knowledge and inability to explain implementation decisions. Misunderstanding of system architecture and technical fundamentals.	System successfully passes basic/core test cases (minimum 50% of provided test cases). Basic grasp of technical concepts with some inaccuracy in explanations. Can demonstrate working functionality but limited understanding of underlying technical implementation. Awareness of system components but superficial technical explanations.	System passes comprehensive range of test cases (50-70% of provided test cases). Clear technical explanations with good understanding of implementation choices. Demonstrates solid grasp of system architecture and can explain technical decisions in wider context. Good awareness of technology implications and trade-offs	System passes all or nearly all provided test cases (70%+ of provided test cases) with robust error handling. Comprehensive technical understanding with clear explanations of implementation limits and opportunities. Demonstrates mastery of system architecture with optimal technical solutions. Excellent awareness of technology implications, trade-offs, and potential system improvements.
Database Operations & Containerisation (20%) **Relates to MO2 and MO3**	No use or unsuccessful use of database and model classes. Misunderstanding of database fundamentals. No use of containerisation.	Partially implemented database and model classes with some form and template support. Some CRUD operations functioning. Basic awareness of database technology. Basic use of containers (less than 2 containers).	Successful use of database and model classes with forms and templates for all CRUD operations. Clear explanations in wider technical context. Awareness of limitations. Successful containerisation (database and web app distributed).	Optimal database use well-embedded in model and form classes. Results successfully displayed via viewer classes and templates. Connection managed efficiently at application scope. Optimal containerisation (database, app, web separated).
Authentication & Authorisation (20%)	No authentication developed or unsuccessful implementation. No awareness of security fundamentals.	Login function developed but not successfully integrated. Login function developed but not successfully integrated.	Successful development and use of authentication and authorisation. Clear understanding of security knowledge applied.	Comfortable use of authentication and authorisation. Comprehensive security knowledge demonstrated.

Relates to MO2 and MO4			Awareness of limitations and implications.	Clear awareness of limitations and implications.
External Services Integration (10%) **Relates to MO2 and MO4**	No or unsuccessful use of external services. No awareness of service concepts.	Basic understanding of service concepts. Partial success in development. Connection errors are not handled properly. Limited understanding of implications.	Successful development and use of services. Partial correctness in business logic implementation. Good understanding of concepts and limitations.	Comfortable development and use of services for intended purpose. Comprehensive understanding of concepts. Clear awareness of limitations and implications.

Appendices

No content.